ANX-PR/CL/001-01
LEARNING GUIDE

SUBJECT

303000093 - Álgebra Avanzada

DEGREE PROGRAMME

30AH - Máster Universitario En Matemáticas Avanzadas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

Index

Learning guide

1. Description

1.1. Subject details

Name of the subject	303000093 - álgebra Avanzada
No of credits	6 ECTS
Type	Optional/elective
Academic year ot the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	30AH - Máster Universitario en Matemáticas Avanzadas
Centre	30 - Esc. Politéc. Enseñanza Superior (epes)
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Alfonso Zamora Saiz (Subject coordinator)		alfonso.zamora@upm.es	Sin horario. Contact by e-mail

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Knowledge about groups, rings and fields, usually covered in courses Algebraic Structures and Algebraic Equations

4. Skills and learning outcomes *

4.1. Skills to be learned

CO1 - Conocer los principales métodos y teorías matemáticas, próximos a la frontera del conocimiento, que permitan tener una amplia perspectiva en al menos tres de las siguientes áreas: Análisis Matemático, Métodos Numéricos, Ecuaciones Diferenciales, Estadística y Ciencia de Datos, Álgebra, Geometría y Topología. TIPO: Conocimientos o contenidos.

CO2 - Conocer demostraciones matemáticas avanzadas que permitan relacionar de una manera rigurosa las hipótesis con las conclusiones en al menos tres de las áreas relacionadas en el CO1.TIPO: Conocimientos o contenidos.

CO3 - Conocer diversas aplicaciones de las teorías matemáticas y su modelización en otros entornos o disciplinas que permitan adaptar los conocimientos matemáticos a nuevos problemas.TIPO: Conocimientos o contenidos.

CP1 - Utilizar programas informáticos especializados como laboratorio para diseñar, construir o validar hipótesis científicas en.alguno de los ámbitos de las matemáticas. TIPO: Competencias.

CP2 - Ser capaz de trabajar en equipos multidisciplinares que involucren conocimientos matemáticos avanzados en áreas teóricas o aplicadas.TIPO: Competencias.

CP3 - Seleccionar y comprender la bibliografía científica sobre un tema concreto, extrayendo la información relevante. TIPO:Competencias.

CP4 - Comunicar de forma oral y escrita conocimientos y conclusiones científicas, valorando diferentes interpretaciones en públicos especializados y no especializados de una manera clara y rigurosa. TIPO: Competencias

CP5 - Complementar la formación matemática de manera autónoma en avances específicos de las Matemáticas, seleccionando la bibliografía adecuada y planteando cuestiones de interés. TIPO: Competencias

CP6 - Reflexionar sobre la relevancia social y ética de la aplicación de los conocimientos adquiridos a determinados problemas.TIPO: Competencias

HA1 - Aplicar los conocimientos matemáticos adquiridos a la resolución de problemas nuevos de una manera crítica, generando hipótesis de trabajo claras, definiciones precisas, nuevos enfoques con fundamento matemático y soluciones originales. TIPO: Habilidades o destrezas

HA2 - Ser capaz de abstraer y detectar las dificultades relevantes ante problemas complejos para poder plantearlos correctamente y resolverlos con técnicas matemáticas avanzadas. TIPO: Habilidades o destrezas

HA3 - Dominar el rigor matemático en las técnicas de demostración aprendidas para su posterior uso en el enunciado y demostraciones de nuevos teoremas. TIPO: Habilidades o destrezas

HA4 - Poseer destreza para proponer, validar e interpretar modelos matemáticos de diferente dificultad en situaciones reales complejas, utilizando las herramientas matemáticas más adecuadas a los fines que se persigan, y valorando las hipótesis y conclusiones asociadas a cada nivel de dificultad del modelo. TIPO: Habilidades o destrezas

HA5 - Adaptar teorías y técnicas matemáticas avanzadas al estudio de problemas en otros ámbitos con criterio científico para valorar su repercusión y formular las hipótesis adecuadas. TIPO: Habilidades o destrezas

HA6 - Interpretar adecuadamente la relevancia de nuevos resultados en Matemáticas, su aplicabilidad en situaciones particulares donde pudieran suponer nuevos avances y su relación con los resultados existentes. TIPO: Habilidades o destrezas

4.2. Learning outcomes

RA2 - Los resultados de aprendizaje en las tres categorías propuestas: conocimientos/contenidos, habilidades y competencias son los señalados en la memoria del título

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

This a classical course on Commutative Algebra, the abstract algebra which is most useful for other branches of mathematics. The course will formally present all statements and proofs, with a view towards the geometrical meaning of the algebraic notions. The central idea is that studying commutative algebra is equivalent to studying (affine) algebraic geometry, which will do basically by studying the ring of polynomials with coefficients in the ring of integers or in a field. To understand this correspondence between algebraic and geometric notions we will use try to visualize notions such as irreducible, reduced, noetherian, smooth or singular, as well as cover localization and completion as the algebraic way to approach the local and global geometry of an algebraic variety. The goal is to place students in the starting point to further study algebraic and arithmetic geometry and number theory.

We will follow basically the text *M. Reid, Undergraduate Commutative Algebra, Cambridge University Press 1995* as a reading course, where students read the indicated sections before class and the instructor develops the concepts and proofs, plus some examples and exercises. A list of problems will be extracted from this text and *M.F. Atiyah, I.G. Macdonald, Introduction to commutative algebra, Addison- Wesley Publishing Co., 1969,* constituting student's homework to acquire the knowledge and skills to face the final exam. Extra topics which cannot be covered in the course because of time will be assigned as projects to groups of two students who will write a paper and give a presentation to the class.

5.2. Syllabus

1. Rings and Ideals.
   1.1. Review of ring theory. Prime and maximal ideals. Zero divisors, nilpotents, radical.
   1.2. Spectrum of a ring. Examples and geometric interpretation.
   1.3. Local rings.
2. Modules
   2.1. Definitions. Homomorphisms of modules. Generators. Free modules.
   2.2. Cayley-Hamilton theorem. Narakama's lemma.
   2.3. Basics on categories and functors. Exact sequences.
3. Noetherian rings and modules
   3.1. Ascending chain condition. Noetheriand rings. Noetherian modules.
   3.2. Hilbert's basis theorem.
4. Finite extensions and integral dependence
   4.1. Finite and integral algebras. Integral closure.
   4.2. Normality. Singularity. Noether's normalization lemma. Weak Hilbert's Nullstellensatz.
5. Geometry of the spectrum of a ring
   5.1. Correspondence between points and ideals.
   5.2. Hilbert's Nullstellensatz.
   5.3. Zariski topology. Algebraic varieties.
6. Rings of fractions and localization
   6.1. Rings of fractions. Localization in rings.
   6.2. Modules of fractions. Localization in modules.
7. Primary decomposition
   7.1. Support. Annihilator. Associated primes.
   7.2. Primary ideals. Primary decomposition.
8. Discrete valuation rings
   8.1. Discrete valuation rings. Normal integral domains.
   8.2. Dedekind domains and uniqueness of factorization. Resolution of singularities.

9. Completion

   9.1. Topologies and completions. Graded rings and modules. Artin-Rees lemma.

10. Dimension theory

   10.1. Hilbert functions. Local regular rings. Transcendent dimension.

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
2	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
3	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
4	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
5	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
6	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			

7	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
8	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
9	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
10	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
11	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
12	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
13	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			
14	**Theory and examples** Duration: 03:00 Lecture **Problem session** Duration: 01:00 Problem-solving class			

15	**Theory and examples** Duration: 01:00 Lecture **Problem session** Duration: 01:00 Problem-solving class **Project presentations** Duration: 02:00 Additional activities			**Attendance and participation in class** Other assessment Progressive assessment and Global Examination Not Presential Duration: 00:00 **Homework problems** Individual work Progressive assessment and Global Examination Not Presential Duration: 00:00 **Project (short note + public presentation)** Group work Progressive assessment and Global Examination Presential Duration: 02:00
16				
17				**Final exam (theory + problems)** Written test Progressive assessment and Global Examination Presential Duration: 03:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
15	Attendance and participation in class	Other assessment	No Presential	00:00	10%	/ 10	CO1 CO3 CP2 CP6 HA1 HA4 CP3 CP1 HA2 HA3 CP4 HA5 CP5 HA6 CO2
15	Homework problems	Individual work	No Presential	00:00	15%	4 / 10	CO1 CO3 CP2 CP6 HA1 HA4 CP3 CP1 HA2 HA3 CP5 HA6 CO2 CP4 HA5
15	Project (short note + public presentation)	Group work	Face-to-face	02:00	15%	/ 10	CO1 CO3 CP2 CP6 HA1 HA4 CP3 CP1 HA2 HA3

							CP5 HA6 CO2 CP4 HA5
17	Final exam (theory + problems)	Written test	Face-to-face	03:00	60%	4 / 10	CO1 CO3 CP2 CP6 HA1 HA4 CP3 CP1 HA2 HA3 CP5 HA6 CO2 CP4 HA5

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
15	Attendance and participation in class	Other assessment	No Presential	00:00	10%	/ 10	CO1 CO3 CP2 CP6 HA1 HA4 CP3 CP1 HA2 HA3 CP4 HA5 CP5 HA6 CO2
15	Homework problems	Individual work	No Presential	00:00	15%	4 / 10	CO1 CO3 CP2 CP6 HA1 HA4 CP3 CP1 HA2 HA3 CP5 HA6 CO2 CP4

							HA5
15	Project (short note + public presentation)	Group work	Face-to-face	02:00	15%	/ 10	CO1 CO3 CP2 CP6 HA1 HA4 CP3 CP1 HA2 HA3 CP5 HA6 CO2 CP4 HA5
17	Final exam (theory + problems)	Written test	Face-to-face	03:00	60%	4 / 10	CO1 CO3 CP2 CP6 HA1 HA4 CP3 CP1 HA2 HA3 CP5 HA6 CO2 CP4 HA5

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Exam (theory + problems)	Written test	Face-to-face	03:00	100%	5 / 10	CO1 CO3 CP2 CP6 HA1 HA4 CP3 CP1 HA2 HA3 CP4 HA5 CP5 HA6 CO2

7.2. Assessment criteria

To pass this course, students have to get a score greater of equal than 5 by summing up the following weighted items:

- Attendance and participation in class (10%). This requires attending class regularly and actively, plus reading the materials before and after class.

- Homework problems (15%). Threshold 4/10. A list of problems will be given for the students to work on them weekly; a selection of these will be handed over to be graded.

- Project (short note + public presentation) (15%). In pairs, an extra course topic will be assigned to prepare a short note and an oral presentation to the class at the end of the semester.

- Final exam (theory + problems) (60%). Threshold 4/10.

For those who do not pass the course in the ordinary evaluation, there will be a Final extraordinary exam (theory + problems) (100%), as an extraordinary evaluation.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
M. Reid, Undergraduate Commutative Algebra, Cambridge University Press 1995.	Bibliography	Course reading
M.F. Atiyah, I.G. Macdonald, Introduction to commutative algebra, Addison- Wesley Publishing Co., 1969.	Bibliography	
David Eisenbud, Commutative Algebra with a View Toward Algebraic Geometry	Bibliography	
H. Matsumura, Commutative algebra, Benjamin/Cumming co., 1981.	Bibliography	
O. Zariski, P. Samuel, Commutative Algebra, Vols. I and II, van Nostrand 1965.	Bibliography	
D. Cox, J. Little, D. O?Shea, Ideals, Varieties and Algorithms, Springer-Verlag 1992.	Bibliography	
E. Arrondo, A geometric introduction to commutative algebra	Bibliography	Notes in https://blogs.mat.ucm.es/arrondo